



Face recognition application in healthcare using computer web camera

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9.1 Introduction

Digital data and image processing is a broad and promising field that is used predominantly in medicine, industry, space science, press, technology, and science and is developing day by day. With the development and widespread use of technology, computer, and telephone technologies, which are

the blessings of technology that many of us can easily access today, keep up with the intensity and applicability of this development and research [1]. The recognition and digital processing of the human face are based on image processing techniques. Considering that technology is developing rapidly, speed, security, reliability, and ease of use are the most important elements that everyone wants to reach. Face recognition systems are used for personnel entry and exit control of large commercial companies and identification of credit card users, and are widely used in two main areas, such as the detection of legal problems, crime detection, person recognition, driver's license, and passport. If we look at the issue of identity applications in its entirety, the biggest purpose of using a face recognition system is security [2].

One of the important innovations in the development and integration of computers, technology, and security systems is image processing. The image processing technique is used to extract useful information from an image and interpret it when needed. The image to be processed can be obtained with the help of a camera or an optical scanner. To apply different processes to the image and to interpret the images in a meaningful way, different processes should be applied to the images by digitizing them. Image processing techniques are widely used in many fields, such as medicine, military, and industry. Image processing techniques are preferred in applications, such as fingerprint, iris, and face recognition for security purposes. To comment on the image taken with the aid of a scanner or camera, the images must be subjected to a processing sequence [3].



9.2 Literature review

Face recognition software is widely accepted in most industries and in medical applications, especially in the diagnosis and monitoring of diseases. Its application in various industries is estimated to be over 50 years now. Between 1964 and 1966, a team led by Woodrow Bledsoe researched on computer programming for recognizing human faces to solve problems. The team led used a scanner to map human hairline, eye and eye socket, and nose. In this study, the task of the computer was to capture this human match, but unfortunately the team was not successful. In Woodrow Bledsoe's words: "The problem of facial recognition is getting harder due to the large variation in the rotation and tilt of the head,

light intensity and angle, facial expression, aging.” The solution to these problems, i.e., the development of camera technology, face mapping processes, computer and machine learning, and processing speeds, have surpassed the age of face recognition. The failure of the study led scientists to new discoveries and search for new technologies [4].

With the invention and development of 2D technology, many camera systems create a flat image of a face and map points such as eye dimensions, nose, and cheekbones, which are called nodes, then the system calculates the relative position of the nodes and converts the data into a digital code. Recognition algorithms search a registered database of faces for a match. 2D camera technology systems give the best results in stable and well-lit conditions, but in dark areas, this success is severely affected. In addition, it is important that objects remain as still and stationary as possible. Unfortunately, the biggest deficit of this system is deceiving the system with photography [5]. One way to overcome the system’s flaws to prevent the 2D system from being deceived by a still photograph was to detect vitality. These systems will look for indicators of a nonlive image, such as inconsistent features between the foreground and background in the image. They may ask the user to blink or move while capturing images. With the detection of vitality, criminals trying to deceive facial recognition systems by using photographs or masks can be prevented. Another revolution in facial systems is the deep convolutional neural network [6]. A convolutional neural network is applied in the detection of image patterns of the model by the user’s machine or computers. The system got this name because it uses an artificial neuron network that mimics the functioning of the human brain, and in reality, this network acts like a black box. It gives the results as yet unknown input values, and then checks to make sure the network is delivering the expected result [7]. If there is a difference or problem in the result or situation, the system will make the adjustments again until it is configured properly and will be able to produce the expected outputs systematically. Today, previous advanced processes are evolving into mass-market devices. The technology is used in many fields today and one of the latest developing technologies is 3D technology. For example, 3D camera technology is used to power the thermal infrared-based face ID feature in Apple’s iPhone X, which many people use today. Thermal IR imaging provides images by mapping facial models, mainly from the model of superficial blood vessels under the skin [8].

There are two types of face recognition systems: static and dynamic. While the static system creates a person recognition system from controlled and regular photographs, the dynamic system obtains image information from a video that is uncontrolled and has a fast data flow. It is possible to get the most accurate results by adjusting brightness settings, suitable filters, and threshold value settings in the systems.

While we are defining people, we can count many characteristics as personality traits. These characteristics can be given as examples of the person's age, weight, height, and eye color, and hair color, but these features can be changed quickly with safety in mind [9]. In addition, it is almost impossible for human memory to fully remember all the features it has seen, and the possibility of having more than one person with the same characteristics is another point of view. The importance of using the face recognition system becomes even more important at this point because the most prominent parts of a person's body, namely the eyes, nose, mouth, and eyebrows, cannot change without a doctor's intervention [10].

The research titled *Impaired face-like object recognition in premanifest Huntington's disease*, by Martínez-Horta et al. [11], introduces the audience to the application of face recognition objects to diagnose the presence of Huntington's disease. They applied a face-like object during an electroencephalographic process and studied on 170 components of face-like objects. The research, however, revealed increasing differences in the selected face-like objects [12–14].

9.2.1 Face recognition system algorithms

The system that identifies each person's face with a mathematical algorithm and an unchangeable and nonrepeatable code during the face recognition process, is designed to maximize the accuracy of the human face by processing the algorithm of points, such as lips and nose. The faces can be searched for matching in historical or live recordings in line with the standards determined by the institution, organization, or person using them. The source of the scanned images is not necessarily images from security cameras alone. Videos, pictures, or animated simulations on platforms, such as smartphones, cameras, or YouTube can be uploaded to the system and scanned [15]. Face recognition software examines the faces in these videos, pictures, and simulations with its advanced algorithm. If there is a blacklist prepared by the institution, organization, or person to be matched, it can detect the people on this list and give a warning. The

accuracy of the solution is that it is a learning software running in the background. The system, where the camera images of the same person can be matched from different angles, can prevent erroneous selections and warnings in subsequent scans. To further increase accuracy, users can manually add contacts to the system [16]. The more images from different sources specific to the relevant person are manually uploaded to the system, the higher the learning probability and correct identification performance of the software. Ideal for places with heavy traffic such as shopping malls and stadiums, factories, and facilities with high-security levels, these systems are preparing to become an indispensable element of smart city applications that will increase in number in the near future [17].

9.2.2 Application of face recognition system

The usage areas of face recognition systems are quite wide. Today, this system, which is mostly used in airports, customs police checkpoints, and places where there is a high density of people, has the feature of being shaped according to the purpose of use. Face recognition systems can be shown as examples in schools, factories, public institutions and organizations, hospitals, shopping centers, plazas, military areas, airports, municipalities, and places with a high density of people. It is frequently used today for staff training. It is possible to trigger turnstiles, gates, and barriers with face recognition systems. [18–22].



9.3 Methodology

The research work proposes to design a Personal Disease Information Access System with a web camera using the patient's picture as the research goal. The study used the Python program and other program libraries such as OS, Numpy, Pandas, face recognition, Selenium, OpenCV, DateTime, and Matplotlib. Similarly, a website builder named YOLA was used to allow data streaming on the internet [22–24]. In addition, criteria such as age, height, weight, date of birth, infectious disease, and smoking habits were used to search on the databases for analysis. Furthermore, the following steps were followed in the research:

1. Create a specific person storage area and upload the photographs and personal information of the target audience to the storage area.
2. Determine whether they have personal identity information, Covid-19, and other health-related data stored on the system.

9.3.1 Some of the explored libraries

OS: The OS module is a module that comes ready-made in Python, allowing us to easily perform operations on files and directories. Considering that the operating systems of different operating systems are different from each other, the OS module offers many functions that enable us to communicate with different operating systems in a consistent manner.

Numpy: Numpy is a math library that allows us to do scientific calculations quickly. Also, unlike Python lists, Numpy arrays must be homogeneous, i.e., all elements in the array must be of the same data type.

Pandas: Pandas is one of the open-source Python libraries that offer easy-to-use data structures and enable data analysis. Due to its many features, it provides great convenience to those working in the field of machine learning in the data reading and preprocessing stages.

Face_Recognition: It is a library to recognize maps of the faces in the code.

Selenium: Selenium is a tool that allows you to open a browser, such as Chrome or Firefox, on the screen with the help of a driver you install on your computer and run all the processes you want with the help of a programming language like a real person.

OpenCV: It is an open-source image processing library written to create a common infrastructure for computer vision applications developed by Intel.

DateTime: It consists of some classes that offer us various functions and attributes so that we can do operations based on time and date. There are three different classes within this module.

Matplotlib: It is used to construct pictures by using numerical values.

The system designed using these libraries automatically enters the website created for this research and obtains the required information, as shown in [Fig. 9.1](#).

From the instructions in [Fig. 9.1](#), the `def getpatient()` function is defined to connect this interface to the designed website using the instruction number. Since the browser dimension is small, we increased these dimensions to the maximum level using instruction line number 18. We typed the names of the people we entered on our website in H5 font; the determining element to call those names. We wrote the contact information in P font on our website. These instructions helped in pulling the information. We learned the selector of H5 fonts with the review button on the website and saved it as people. We did the same for the P font and introduced it to our program as information. We created a space to store the information that comes in rows 24, 25, 26, and 27.

```

12
13
14
15 def getpatient():
16
17     driver.get("https://tevmusyavgen.yolasite.com/")
18     driver.maximize_window()
19
20
21     people = driver.find_elements_by_css_selector(".ws-section.ws-surface.ws-dark-1")
22     informations = driver.find_elements_by_css_selector(".ws-section.ws-surface.ws-light-0")
23
24     patients = []
25     patientsForDisplaying = []
26     infos = []
27     sex = []
28

```

Figure 9.1 Data information.

```

28
29     for i in people:
30
31         nameComing = i.find_element_by_css_selector('.ws-m-rich-text.ws-m-blockquote-with-icon').text
32         Tr2Eng = str.maketrans("çğıöşüÇĞİÖŞÜ", "cgiosuCGİOSU")
33         nameComing = nameComing.translate(Tr2Eng)
34
35         seperated = nameComing.upper().split('-')
36         justName1 = seperated[0].split()
37
38         patientsForDisplaying.append(seperated[0])
39         patients.append("".join(justName1))
40         sex.append(seperated[1])
41         # print(patients)
42
43     for j in informations:
44         infos.append(j.find_element_by_css_selector('.ws-m-rich-text.ws-m-blockquote-with-icon').text)
45
46     return patients, infos, sex, patientsForDisplaying
47
48
49
50 def texter(name,data,sex):
51     bg = cv2.imread('background.png')
52
53     ndata = data.split('/')
54
55     cv2.putText(bg,name,(40,80),cv2.FONT_HERSHEY_TRIPLEX,0.9,(0, 0, 0),2)
56     cv2.putText(bg,sex,(720,80),cv2.FONT_HERSHEY_TRIPLEX,0.9,(0, 0, 0),2)
57

```

Figure 9.2 Name coming.

As shown in Fig. 9.2, in the 31st row, the web addresses of the names are selected one by one and assigned to the “name coming” for every “i.”

On line 32 and 33, our code converts the Turkish characters seen on our website into English characters. And thus there is no incompatibility.

In the 35th and 36th rows, name information does not come to the code by only names but more. So that we can use all different information

individually, we separate that information. Afterward, the information is assigned to the lists in the 39th and 40th rows. The 43rd and 44th rows do the same thing for patient information. Finally, in the 46th row, the function returns all those valuable data so that they can be used whenever needed.

In the 50th row, the `texter` function is defined by using `def texter()`. In the `texter` function, we display all the information coming from the database.

Fig. 9.3 shows the representation of the selenium library in the database. We used a `WebDriver` to reach the database. To do that we use a `WebDriver`, which is introduced by using the Selenium library in the 70th row.

In the 71st row, we call the `getpatient` function, and all the returns will be assigned to the lists on the left-hand side. As our work with the `WebDriver` is done, we can close it, and it can be seen in the 72nd row.

To capture a video, we call a function from the `cv2` library, and also create a “while loop” to collect the information from the image sources and compare them one after the other with the live picture on the opened camera. The collected data, with the help of these lines of instructions, generated the image in Fig. 9.4.



9.4 Results

The results in Fig. 9.5 shows that by using image sources that we have imported, functions that we have defined, more functions from `cv2`, `Face_Recognition`, `Numpy` libraries, and finally the live camera, it becomes viable to get face detection. In addition to that, when the picture between the camera and the sources matches up, the break loop on the 152nd row starts doing its job and breaks the while loop.

```

58     for i in range(1,len(ndata)+1):
59
60         ndata[i-1] = ndata[i-1].strip()
61         belower = 80*i+80
62         cv2.putText(bg,ndata[i-1],(40,belower),cv2.FONT_HERSHEY_TRIPLEX,0.9,(0, 0, 0),2)
63
64
65     cv2.imshow("window", bg)
66     cv2.waitKey(0)
67
68
69
70     driver = webdriver.Chrome()
71     [patients,infos,sexes,namesToBeDisplayed] = getpatient()
72     driver.close()

```

Figure 9.3 Selenium library.

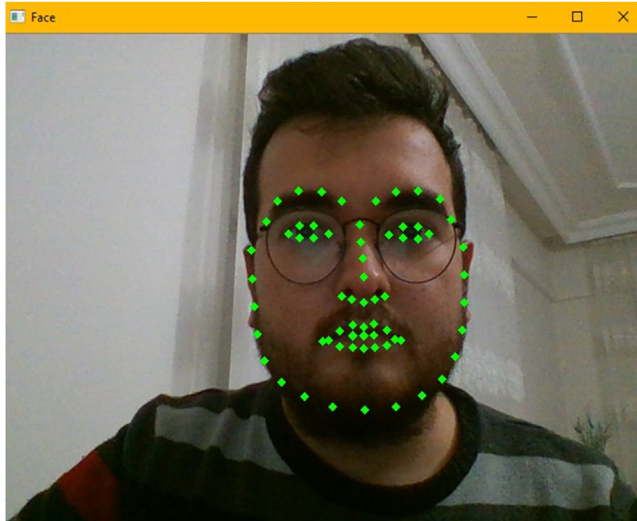


Figure 9.4 Face recognition.

```

165
166 sayac = 0
167
168 for k in range(0,len(patients)):
169     if (patients[k] == name):
170         print(infos[k])
171         break
172     sayac += 1
173
174
175 texter(namesToBeDisplayed[sayac],infos[sayac],sexes[sayac])

```



```

135
136     if matches[matchIndex]:
137         name = classNames[matchIndex].upper()
138         # print(name)
139         y1,x1,y2,x2 = faceLoc
140         y1,x1,y2,x2 = 4*y1, 4*x1, 4*y2, 4*x2
141         cv2.rectangle(img, (x1,y1), (x2,y2), (0,255,0),2)
142         cv2.rectangle(img, (x1,y2-35), (x2,y2), (0,255,0),cv2.FILLED)
143         cv2.putText(img,name,(x1+6,y2-6),cv2.FONT_HERSHEY_COMPLEX,1,(255,255,255),2)
144         # readInfo(name)
145         num += 1
146
147
148     cv2.imshow('Webcam', img)
149     cv2.waitKey(1)
150
151     if num == 2:
152         break
153

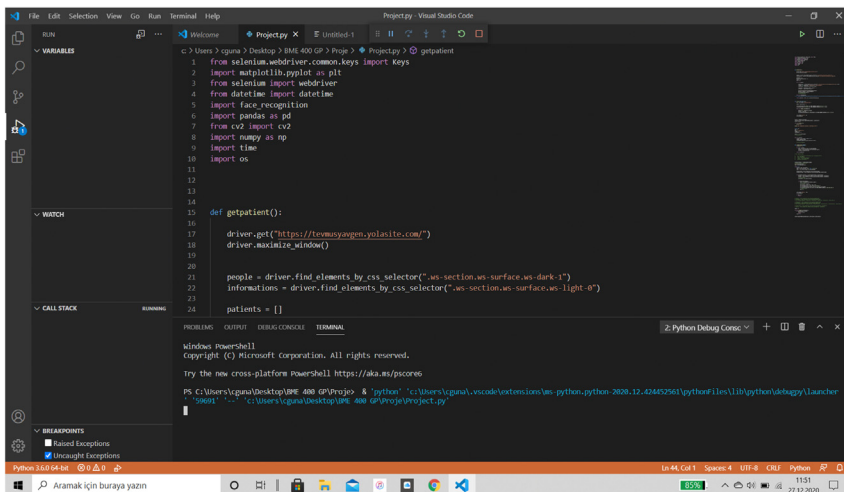
```

Figure 9.5 cv 2.

The camera gets opened when the program starts, and recognizes the person looking at the camera. Now, we need to display this person's information (name, gender, age, blood type, allergenics, etc.) on the screen. Consequently, the “for loop” in the 168th row starts the searching process from the information in the database (there are lots of patients, so we need to find the right person). In the 169th row, the “if loop” asks whether we can find the information of the right person. If the answer is yes, then it breaks the for loop on the 169th row by using the variable we have defined as SAYAC. After the while loop, we have one more row to run, which is the texter function. When we run this function, by definition, it displays the person's information on the screen. Then, the code ends, as represented in Fig. 9.6.

At first we start the process of writing the 'py.\Project.py' command to the terminal for code execution. The result of the executed code is shown in Fig. 9.7–9.10, which displays the information of some of the authors of this paper.

And finally, if the image of our patient on the camera and the image in the database match, the patient information is reflected on the screen waiting for further support from healthcare professionals in emergency situations.



```

c:\Users\lguna\Desktop> python getpatient.py
1 from selenium.webdriver.common.keys import Keys
2 import matplotlib.pyplot as plt
3 from selenium import webdriver
4 from datetime import datetime
5 import face_recognition
6 import pandas as pd
7 from cv2 import cv2
8 import numpy as np
9 import time
10 import os
11
12
13
14
15 def getpatient():
16
17     driver.get("https://temucaygelen.yolanite.com/")
18     driver.maximize_window()
19
20
21     people = driver.find_elements_by_css_selector(".ws-section.ws-surface.ws-dark-1")
22     informations = driver.find_elements_by_css_selector(".ws-section.ws-surface.ws-light-4")
23
24     patients = []

```

Terminal Output:

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\Users\lguna\Desktop> python getpatient.py

```

Figure 9.6 Concluding section of the coding.

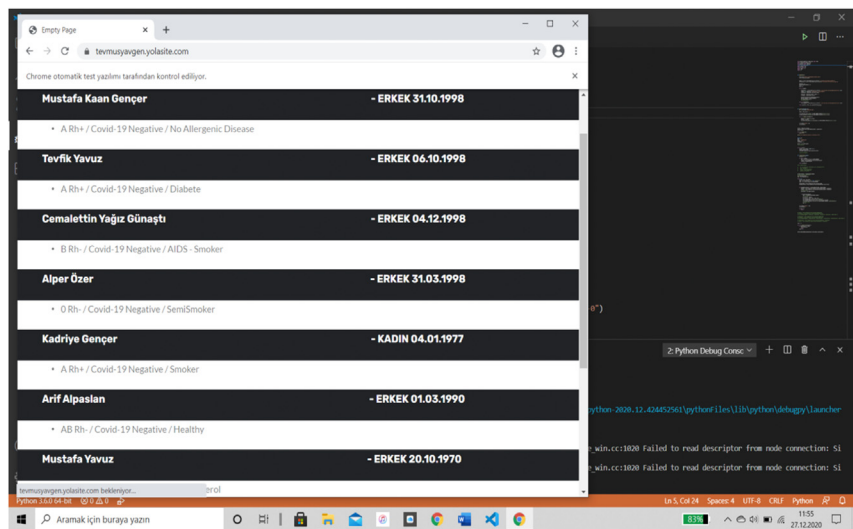


Figure 9.7 Coding results.

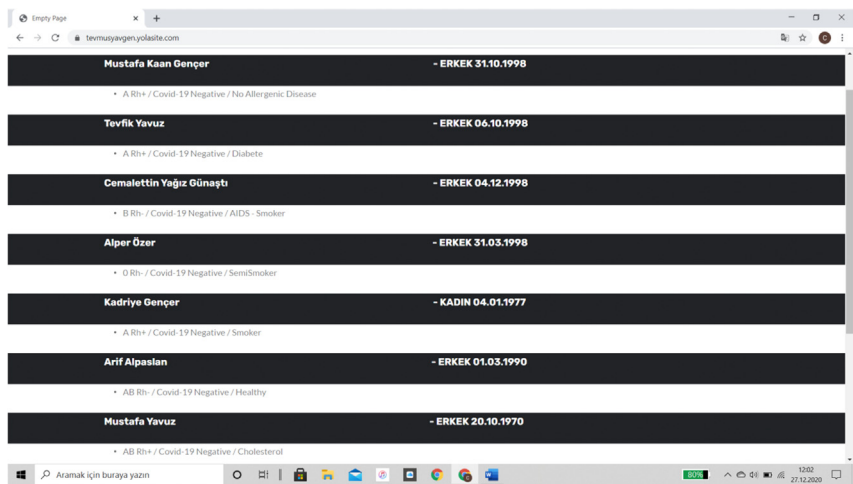


Figure 9.8 Some extracted information.

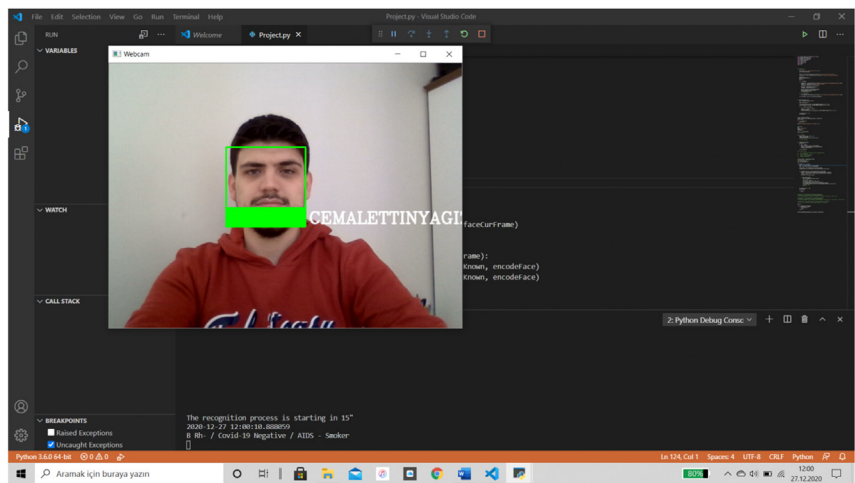


Figure 9.9 More extracted information.

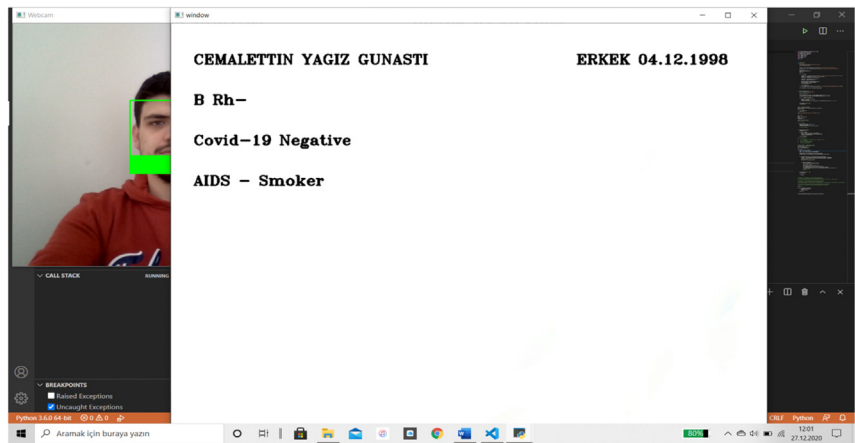


Figure 9.10 Final report.

9.5 Conclusion

This research work focuses on the modern assessment of using novel research methods to handle depressing factors in global health. It focuses on developing self-efficient computer-programmed methods to detect, identify, and get data on patients' well-being using computer programs.

Using the Python program, we chose the priority program libraries installed to call the necessary information and run smoothly. These program libraries are OS, Numpy, Pandas, Face_Recognition, Selenium, OpenCV, DateTime, and Matplotlib. We used a site builder called Yola (Open Access Program), to stream data over the internet. We archived photos of some people with distinct facial features to enable us to identify the facial features of the people we previously captured. We assigned the correct names to the photographs by entering the human information we placed in the folders completely and correctly. We checked the photos in the folder with the photos of people we took both from the internet and the camera. In the field of health systems, the infrastructure of face recognition has not been fully harnessed and this is one of the motivations for this research. In the future, bad experiences by patients can be avoided by advancing this idea and integrating it into the healthcare management systems. This could reduce the turnaround time of doctor-to-patient examination in the hospital. We hope that in the future, the integration of the face recognition system with the healthcare system will be achieved and more efficient healthcare services will be provided to patients.

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